



朱璽 Kaine Zhu



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Github: github.com/zx50416

【學經歷】

國立中山大學

海洋環境及工程學系 學士 (畢業)
2019.6-2023.6

國立中山大學

海洋環境及工程學系 碩士 (畢業)
2023.9-2025.9

國立台灣大學

生物環境系統工程學系 研究助理
2025.9-2025.12

國立中山大學

海洋環境及工程學系 研究助理
2026.1

康橋國際學校

AI 程式設計工程師
2026.7-

【專業技能】

Programming

- Python
- C++
- AI programming (Codex, Claude Code, ...)

Machine Learning & Data Analysis

- Time-Series Forecasting
- Feature Engineering
- Feature Importance Analysis
- Model Training & Evaluation
- Data Preprocessing

System & Data Processing

- Real-Time Data Processing
- API Data Collection & Integration
- Spatial Data Processing
- ETL Pipeline Development
- Automation Script Development
- Forecasting Pipeline Development

【碩士論文 (英文論文)】

Prediction of Deflections on Corrugated Blast Wall under Explosion Loads Using Machine Learning

- 利用數值模擬軟體 (LS-DYNA) 所建立之資料集，以機器學習方法預測波紋鋼防爆牆於爆炸載荷作用下之最大與永久位移。
- 研究內容包含資料前處理、特徵工程、模型驗證及特徵重要性分析，探討結構幾何參數與載荷條件對位移反應之影響。
- 研究期間完成模型建置、參數調校與實驗設計，最終模型於測試資料上取得超過 99% 的預測準確度 ($R^2 > 0.99$)，驗證了機器學習於結構工程問題上的應用價值與可行性。

【研究助理】

- 實現「即時淹水預警系統」模型架構與完整 Python 程式碼編寫。
- 負責專案進度管理，代表研究室與政府機關及外部科技公司進行需求訪談與進度回報。
- 負責將 AI 模型封裝為正式執行檔，並設計數據傳輸邏輯，協助合作廠商將模型預測結果介接至政府公共網站平台。
- 分析資料之潛在模式與特徵分群結果，作為後續預測模型優化與設計之參考。

【其他經驗】

- 具備 ESG、能源與氣候資料相關之 Machine Learning 專案與競賽經驗。
- kaggle: Large-scale Energy Anomaly Detection
- T-Brain: 根據區域微氣候資料預測發電量競賽
- 在學期間擔任室內樂社社長，將前年不理想之社團評鑑成績提升為前 15 名之社團，獲 20000 元之獎學金
- 曾任社團評鑑執行委員一年
- 多次於音樂會中擔任總負責人、指揮、作曲、器樂演出並發行個人專輯，師事國立中山大學音樂系陳以軒教授。



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【MASTER'S THESIS】

Prediction of Deflections on Corrugated Blast Walls under Explosion Loads Using Machine Learning

- Used machine learning methods to predict the maximum and permanent deflections of corrugated blast walls under explosion loads based on datasets generated by LS-DYNA simulations.
- The research included data preprocessing, feature engineering, model validation, and feature importance analysis to investigate the effects of structural geometric parameters and loading conditions on deflection responses.
- During the study, machine learning models were developed, tuned, and evaluated through experimental design. The final models achieved over 99% prediction accuracy ($R^2 > 0.99$) on testing datasets, demonstrating the feasibility and practical value of applying machine learning to structural engineering problems.

【WORK EXPERIENCE】

Research Assistant

National Taiwan University

Department of Bioenvironmental Systems Engineering
2025.9 – 2025.12

- Developed the architecture and implementation of a real-time flood forecasting system using Python.
- Managed project progress and communicated with government agencies and external technology companies regarding project requirements and updates.
- Packaged AI models into deployable applications and designed data transmission workflows to support integration with government public service platforms.

Research Assistant

National Sun Yat-sen University

Department of Marine Environment and Engineering
2026.1

- Analyzed data patterns and feature relationships to support model improvement.

AI Software Engineer

Kang Chiao International School

Department of Marine Environment and Engineering
2026.7 –

【ADDITIONAL EXPERIENCE】

Experience in Machine Learning projects related to ESG, energy systems, and climate data analysis.

- Kaggle Competition: Large-scale Energy Anomaly Detection.
- T-Brain Competition: Solar Power Generation Prediction Using Regional Microclimate Data.

Leadership & Activities

- President of the Chamber Music Club at National Sun Yat-sen University.
- Led the club to improve its annual evaluation ranking into the top 15 student organizations and received a scholarship award of NT\$20,000.
- Organized and led multiple music performances as a project coordinator, conductor, composer, and instrumental performer, and released a personal music album under the guidance of Prof. Yi-Hsuan Chen from National Sun Yat-sen University.

【TECHNICAL SKILLS】

Programming

- Python
- C++
- AI programming
(Codex, Claude Code, ...)

System & Data Processing

- Real-Time Data Processing
- API Data Collection & Integration
- Spatial Data Processing
- ETL Pipeline Development
- Automation Script Development
- Forecasting Pipeline Development

Machine Learning & Data Analysis

- Time-Series Forecasting
- Feature Engineering
- Feature Importance Analysis
- Model Training & Evaluation
- Data Preprocessing

【EDUCATION】

National Sun Yat-sen University

B.S. in Marine Environment and Engineering
2019.9 – 2023.6

National Sun Yat-sen University

M.S. in Marine Environment and Engineering
2023.9 – 2025.9